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Topic 8: Exchange and Transport in Animals

Edexcel GCSE (9-1) Biology (1BI0) — Paper 2

All organisms must take in useful substances (oxygen, glucose, water, ions) and get rid of waste products (carbon dioxide, urea). In a single-celled organism these substances can simply diffuse in and out across the cell surface membrane. But as organisms get bigger, the distances become too great and the surface area is not big enough. This topic explains why multicellular animals need dedicated **exchange surfaces** (like the alveoli in the lungs) and **transport systems** (the blood and the heart), and how these structures are adapted for their functions.

8.1 Surface-Area-to-Volume Ratio (SA:V)

> The **surface-area-to-volume ratio (SA:V)** is the surface area of an object divided by its volume. As an object grows larger, its volume increases faster than its surface area, so SA:V *decreases*.

For a single-celled organism like an amoeba, the cell is small enough that oxygen can diffuse in and CO_2 can diffuse out across the cell surface fast enough to meet its needs. But as organisms get bigger, there is simply not enough surface for diffusion to supply every cell. A large animal therefore needs specialised **exchange surfaces** (lungs, intestine) to take substances in, and a **transport system** (blood and heart) to carry them around.

Worked example: cube SA:V

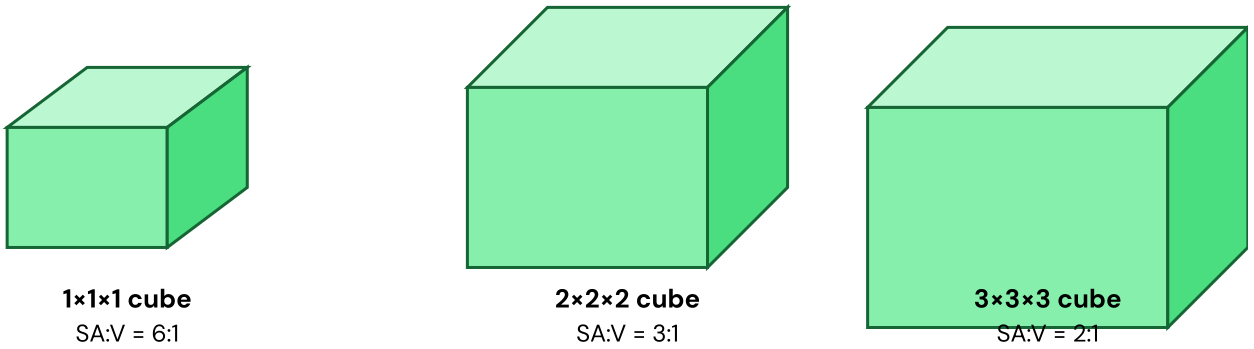
For a cube of side length L: surface area = $6 \times L^2$, volume = L^3, so SA:V = $6/L$.

Side (cm)	Surface area (cm²)	Volume (cm³)	SA:V
1	6	1	6 : 1

2	24	8	3 : 1
3	54	27	2 : 1
4	96	64	1.5 : 1
5	150	125	1.2 : 1

Notice how doubling the side length (from 1 to 2) halves the SA:V. A small organism therefore has a much larger SA:V than a large organism of the same shape.

SA:V decreases as objects get larger



Exam Tip: Always write SA:V as a ratio (e.g. "6:1") or as a number with "per unit volume". If asked "compare", say both values and state **which is bigger and by how much**.

8.2 Why Multicellular Organisms Need Transport Systems

In a small organism, every cell is close enough to the surface for simple diffusion to supply oxygen and remove waste. In a large animal two problems arise:

- **Diffusion distances are too large** — it would take far too long for oxygen to diffuse from the skin to a muscle cell deep inside the body.

- **SA:V is too small** — the outer surface is not large enough to supply the enormous number of inner cells.
- **Energy demands are high** — active tissues (muscle, brain) need a constant, rapid supply of oxygen and glucose, which diffusion alone cannot deliver.

Multicellular animals solve these problems with two linked adaptations:

1. Dedicated **exchange surfaces** (e.g. lungs, small intestine) where substances move between the body and the environment.
2. A **mass transport system** (the circulatory system) that carries substances between exchange surfaces and every cell in the body.

8.3 Adaptations of Exchange Surfaces

All efficient exchange surfaces share the same four features, because each of them increases the rate of diffusion (Fick's law: rate ∝ surface area × concentration difference / distance).

Adaptation	Why it helps	Example in the body
Large surface area	More molecules can cross per second	Folded alveoli in lungs; villi and microvilli in small intestine
Thin walls (short diffusion path)	Particles have less distance to travel, so diffuse faster	Alveolus wall one cell thick; capillary wall one cell thick
Good blood supply	Carries substances away / brings them in, keeping a steep concentration gradient	Dense capillary network around alveoli and villi
Ventilation or flow of medium	Maintains a steep concentration gradient on the other side	Breathing refreshes air in the alveoli; water flowing over fish gills

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Exam Tip: When asked "explain how this is adapted for gas exchange", always link each feature to an *effect on diffusion rate*. Don't just list features — say *why* they matter.

8.4 Respiration

Respiration is a series of chemical reactions in cells that *release energy from glucose*. The energy is stored temporarily in a molecule called **ATP**, which powers all the energy-requiring processes in the cell. Respiration is *not* breathing — it happens inside every living cell.

Aerobic respiration (with oxygen)

Occurs in the **mitochondria**. Requires oxygen. Releases a large amount of energy per glucose molecule.

```
glucose + oxygen  →  carbon dioxide + water  (+ energy/ATP)

C₆H₁₂O₆ + 6 O₂  →  6 CO₂ + 6 H₂O
```

Anaerobic respiration in humans (no oxygen)

Happens in muscle cells during vigorous exercise when oxygen cannot be supplied fast enough. Glucose is only *partially* broken down, so much less energy is released.

```
glucose  →  lactic acid  (+ a little energy)
```

Lactic acid builds up and lowers the pH of the muscle, causing fatigue and cramp. The oxygen needed after exercise to break down the lactic acid is called the **oxygen debt** — it is why you keep breathing heavily after you stop running.

Anaerobic respiration in yeast (fermentation)

Yeast is a single-celled fungus. In the absence of oxygen, it carries out **fermentation**:

```
glucose  →  ethanol + carbon dioxide  (+ a little energy)
```

This reaction is used in **brewing** (the ethanol produces alcohol) and **breadmaking** (the CO_2 bubbles make the dough rise; the ethanol evaporates during baking).

Comparison table

Feature	Aerobic	Anaerobic (human muscle)	Anaerobic (yeast)
Oxygen needed?	Yes	No	No
Glucose broken down	Completely	Partially	Partially
Products	$CO_2 + H_2O$	Lactic acid	Ethanol + CO_2
Energy (ATP) released per glucose	Large (~38 ATP)	Small (2 ATP)	Small (2 ATP)
Location	Mitochondria	Cytoplasm	Cytoplasm
Uses	Normal cell activity	Sprinting, heavy lifting	Brewing, breadmaking

> **Exam Tip:** Learn the **symbol equation for aerobic respiration by heart** — it often carries a whole 2 marks on its own. Balance check: 6 C + 12 H + 18 O on each side.

8.5 Uses of Energy from Respiration

The energy released by respiration is used for:

- **Muscle contraction** — movement, posture, heartbeat.
- **Active transport** — moving ions/molecules against a concentration gradient (e.g. mineral ion uptake in root hair cells, glucose reabsorption in kidneys).

- **Synthesis of large molecules** — building proteins from amino acids, DNA replication, making starch/glycogen.
- **Keeping warm** — mammals and birds release heat from respiration to maintain a constant body temperature.
- **Cell division (mitosis)** — growth and repair require large amounts of energy.

8.6 Investigating Respiration Rate

Respiration rate can be measured by either the **uptake of oxygen** or the **release of carbon dioxide**.

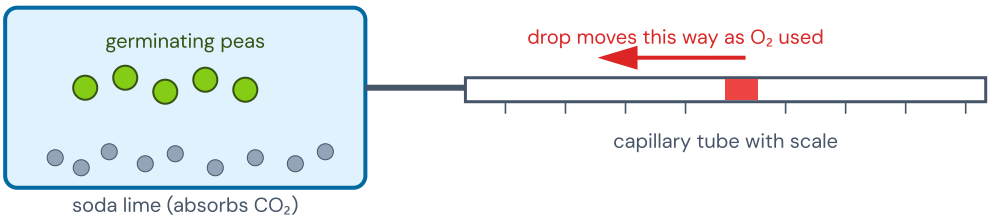
Method 1 — Limewater (detecting CO_2)

Breathed-out air is bubbled through limewater. CO_2 turns limewater from clear to milky/cloudy. Comparing the speed at which the limewater turns cloudy between rest and after exercise shows that exercise increases the rate of respiration and therefore CO_2 production.

Method 2 — Respirometer (measuring O_2 uptake)

1. Place a small organism (e.g. germinating peas or woodlice) in a sealed test tube containing **soda lime** (absorbs all CO_2 produced).
2. Connect a capillary tube containing a coloured liquid drop.
3. As the organism respires, oxygen is taken up but the CO_2 is absorbed, so the total gas volume falls and the drop moves *towards* the organism.
4. The distance moved per minute is a measure of the rate of oxygen consumption (and therefore respiration).

Respirometer



Control variables: temperature (use a water bath), mass of organisms, type of organism.
Control experiment: a tube with dead/boiled organisms, to rule out any changes caused by temperature or air pressure.

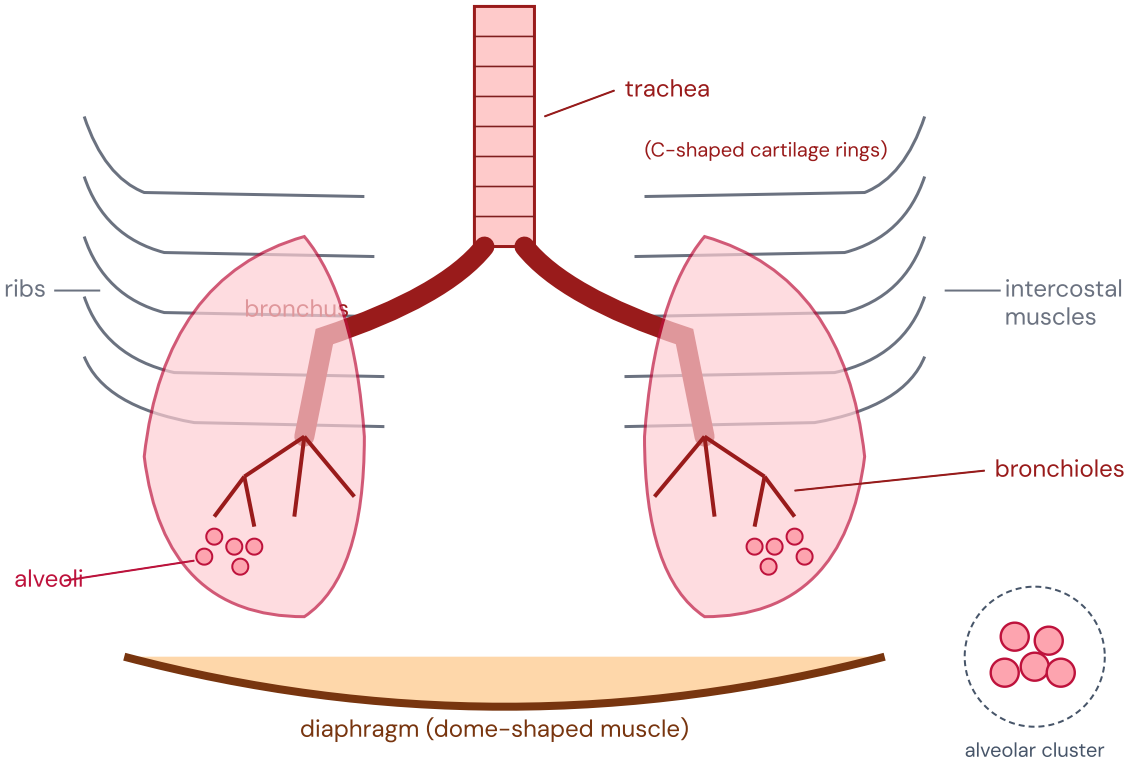
8.7 The Human Gas Exchange System

Air is drawn in through the nose and mouth, passes down the **trachea**, which divides into two **bronchi** (one per lung), then into smaller **bronchioles**, ending in millions of tiny air sacs called **alveoli** where gas exchange with the blood takes place.

The route of air is: **mouth/nose** → **trachea** → **bronchi** → **bronchioles** → **alveoli**.

The trachea is held open by **C-shaped rings of cartilage**. The walls of the bronchioles contain smooth muscle. The **ribs** form a protective cage; the **intercostal muscles** run between the ribs. The **diaphragm** is a dome of muscle below the lungs separating the thorax from the abdomen.

The Human Respiratory System

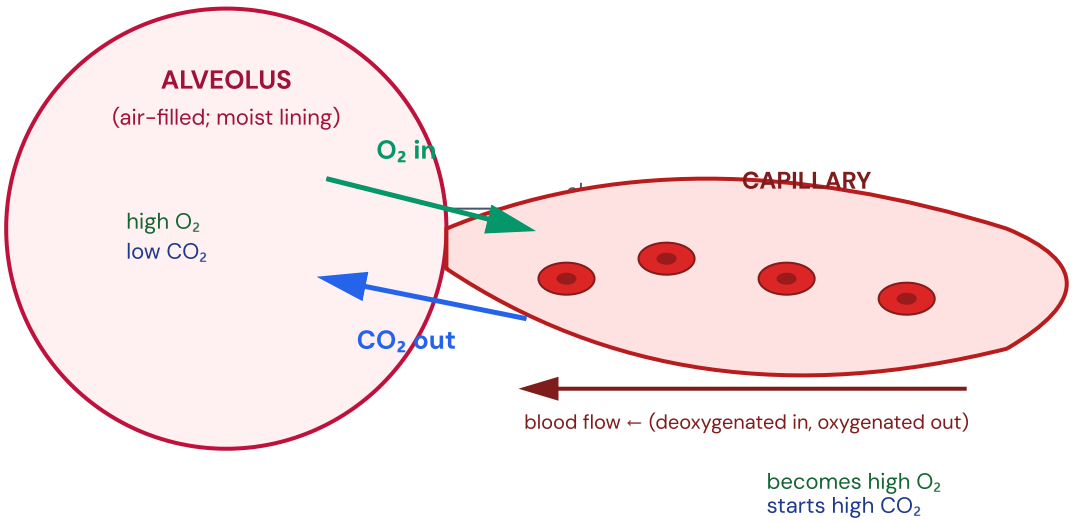


8.8 Adaptations of Alveoli for Gas Exchange

Each lung contains around 300 million alveoli, giving a combined surface area of about **70–100 m²** — roughly the size of a tennis court. Alveoli are *exceptionally* well-adapted exchange surfaces:

Adaptation	How it helps gas exchange
Huge total surface area (~70 m²)	More O_2 and CO_2 can diffuse per second
Walls are only one cell thick	Short diffusion distance between air and blood
Capillary walls are also one cell thick	Blood and air are separated by only two flat cells
Dense capillary network	Keeps concentration gradient steep (constant O_2 removal, CO_2 supply)
Moist inner lining	Oxygen dissolves so it can diffuse through the membrane
Ventilated by breathing	Fresh oxygen-rich air constantly replaces used air

Gas Exchange at an Alveolus



> **Exam Tip:** "Why are alveoli moist?" Answer: so that oxygen can **dissolve** in the water lining before diffusing across the alveolar wall into the blood. Dry membranes cannot exchange gases efficiently.

8.9 Ventilation — Inhalation and Exhalation

Ventilation is the movement of air in and out of the lungs. It is produced by changes in the volume and pressure of the thorax, which follow the principle that *when volume increases, pressure falls — and air flows from high to low pressure*.

Inhalation (breathing in)

- The **diaphragm contracts** and flattens downward.
- The **external intercostal muscles contract**, pulling the ribs up and out.
- The **volume** of the thorax **increases**.
- The **pressure** inside the thorax **decreases** (below atmospheric).
- Air flows *into* the lungs down the pressure gradient.

Exhalation (breathing out)

- The **diaphragm relaxes** and rises back into its dome shape.
- The **external intercostal muscles relax**; the ribs fall down and in under gravity (and under elastic recoil of the lung tissue).
- The **volume** of the thorax **decreases**.
- The **pressure** inside rises (above atmospheric).
- Air flows *out* of the lungs.

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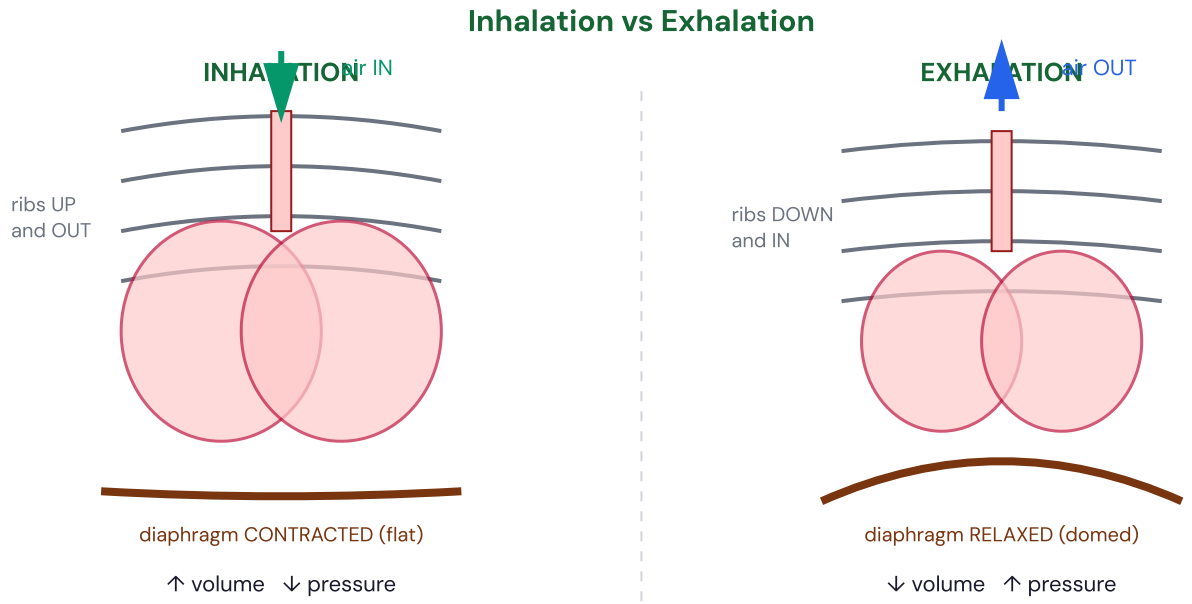

Exam Tip: A common trap: writing "the lungs suck air in". They don't — air is *pushed* in by atmospheric pressure when thorax pressure drops below it. Describe ventilation in terms of volume → pressure → flow.

8.10 Components of Blood

Blood is a tissue made up of four components: three types of cell (and cell fragments) suspended in a yellow liquid called **plasma**.

Component	Description	Main function
Plasma	Straw-coloured liquid; ~55% of blood volume; mostly water	Transports cells, nutrients, CO_2, urea and hormones
Red blood cells (erythrocytes)	Biconcave discs, no nucleus, contain haemoglobin	Transport O_2 from lungs to tissues
White blood cells (leukocytes)	Larger, have a nucleus; phagocytes and lymphocytes	Defence against pathogens

Platelets	Tiny cell fragments, no nucleus	Clotting — seal wounds and prevent pathogen entry

Four Components of Blood

Plasma
(liquid, straw-coloured)

Red blood cell
(biconcave, no nucleus)

White blood cell
(has nucleus)

Platelets
(cell fragments)

8.11 Red Blood Cells

Red blood cells (erythrocytes) are **highly specialised** for transporting oxygen. They are made in the bone marrow and live about 120 days.

Adaptation	Explanation
Biconcave disc shape	Large surface area to volume ratio for faster diffusion of oxygen; flexible enough to squeeze through narrow capillaries
No nucleus	More room to pack in haemoglobin; more oxygen carried per cell
Contain haemoglobin	Iron-containing protein that binds O_2 reversibly
Small and flexible	Can bend to pass single-file through capillaries, bringing O_2 close to every cell

In the lungs (high O_2): haemoglobin + O_2 → **oxyhaemoglobin**. In respiring tissues (low O_2): oxyhaemoglobin → haemoglobin + O_2. This reversible reaction allows red blood cells to pick up oxygen where it is plentiful and release it where it is needed.

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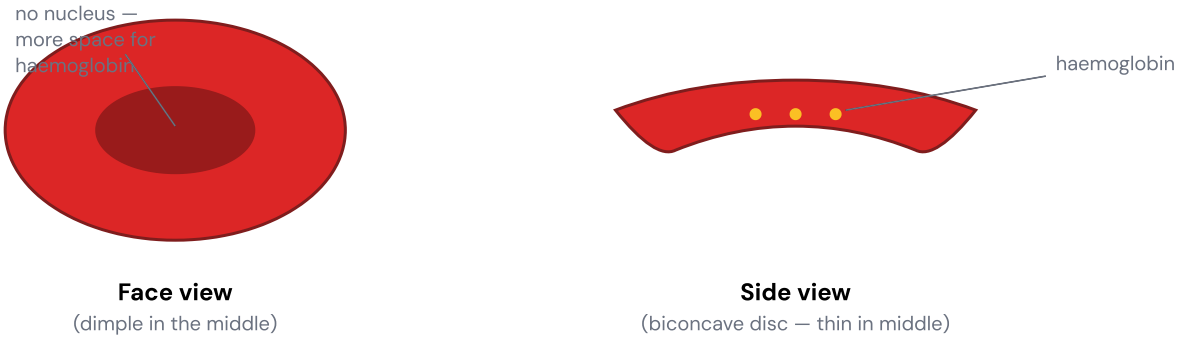
Red Blood Cell — face and side view



8.12 White Blood Cells

White blood cells defend the body against pathogens (bacteria, viruses, fungi). Unlike red blood cells they *have* a nucleus. The two main types are **phagocytes** and **lymphocytes** (covered in more detail in Topic 5).

- **Phagocytes** — engulf and digest pathogens by a process called phagocytosis. They have a multi-lobed nucleus and many lysosomes.
- **Lymphocytes** — produce **antibodies** that bind specifically to a pathogen's antigens. This marks pathogens for destruction and produces immunity (memory cells). Some lymphocytes produce antitoxins to neutralise bacterial toxins.

8.13 Platelets

Platelets are small fragments of cells (they have no nucleus). When a blood vessel is damaged, platelets stick to the wound and release chemicals that trigger a cascade of reactions producing the fibrous protein **fibrin**. Fibrin forms a mesh of threads that traps red blood cells to form a **clot**. The clot:

- Prevents excessive **blood loss**.
- Stops **pathogens** from entering the body.
- Forms a scab under which the skin can heal.

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People who lack clotting factors (e.g. those with **haemophilia**) bleed for longer because clots do not form normally.

8.14 Plasma

Plasma is the straw-coloured liquid that makes up about 55% of blood. It is ~90% water with many substances dissolved in it. Plasma transports:

- Blood cells (red, white, platelets) — held in suspension.
- Dissolved **nutrients** — glucose, amino acids, fatty acids from the small intestine to the tissues.
- **Carbon dioxide** — from respiring tissues to the lungs (mostly as dissolved hydrogencarbonate ions).
- **Hormones** — chemical messengers from glands to target organs.
- **Urea** — from the liver to the kidneys for excretion.
- Heat — moved from active organs (e.g. muscles, liver) to cooler parts of the body.
- Antibodies and blood proteins.

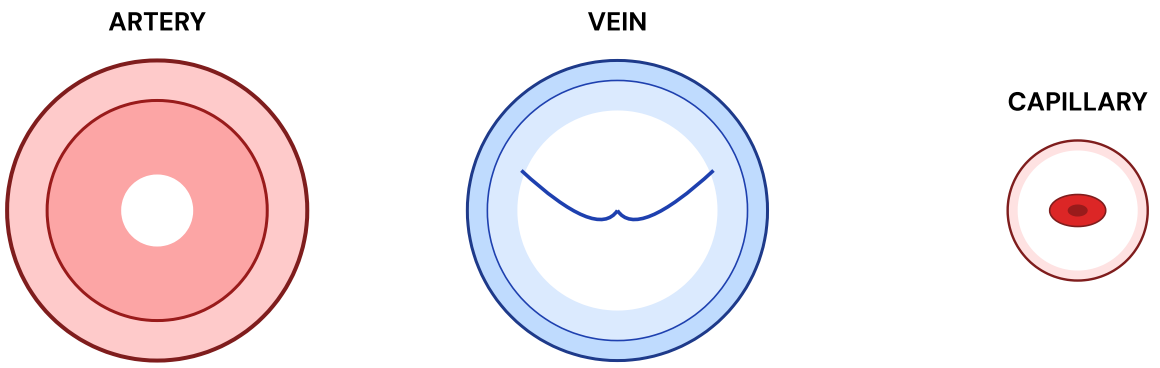
8.15 Blood Vessels

Blood travels through three main types of vessel: **arteries** (carry blood *away from* the heart), **veins** (carry blood *to* the heart), and **capillaries** (thin exchange vessels that link the two). Each is adapted to its function.

Feature	Artery	Vein	Capillary
Direction of flow	Away from heart	Towards heart	Link arterioles to venules through tissues
Pressure inside	High, pulsing	Low, steady	Drops from high to low

Wall	Thick, muscular, elastic	Thin wall	One endothelial cell thick
Lumen	Narrow	Wide	Very small (RBCs pass single-file)
Valves?	No (except at heart)	Yes — prevent backflow	No
Oxygen content	Usually oxygenated (except pulmonary artery)	Usually deoxygenated (except pulmonary vein)	Changes along length (exchange happens here)
Function	Carry high-pressure blood to tissues	Return blood to heart	Exchange O_2, CO_2, nutrients, waste with cells

Cross-sections of Artery, Vein and Capillary



thick muscular wall, narrow lumen, HIGH pressure

thin wall, WIDE lumen, valves prevent backflow, LOW pressure

wall ONE cell thick, tiny lumen, RBCs squeeze through

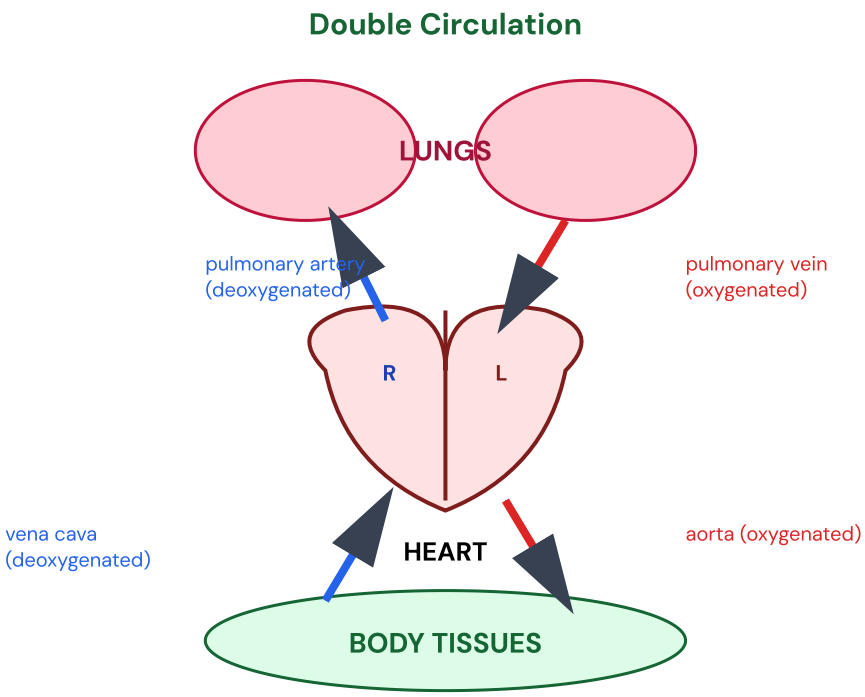
Exam Tip: Remember: **A**rteries carry blood **A**way from the heart. Veins return to the heart (think "v" pointing back). The only *exceptions*: the **pulmonary artery** carries deoxygenated blood, and the **pulmonary vein** carries oxygenated blood.

8.16 Double Circulation

Mammals have a **double circulatory system** — blood passes through the heart **twice** for each complete trip around the body. The heart contains two pumps side by side.

- **Pulmonary circulation** — carries deoxygenated blood from the *right* side of the heart to the lungs and returns oxygenated blood to the left side.
- **Systemic circulation** — carries oxygenated blood from the *left* side of the heart to the body and returns deoxygenated blood to the right side.

Advantage: after being oxygenated in the lungs the blood returns to the heart and is pumped again. This means the blood going to the body leaves the heart at *high pressure*, so it can flow quickly to every tissue. A single circulation (as in fish) cannot do this because blood loses pressure at the gills. Double circulation is essential for supporting the high metabolic rates of mammals and birds.



8.17 The Structure of the Heart

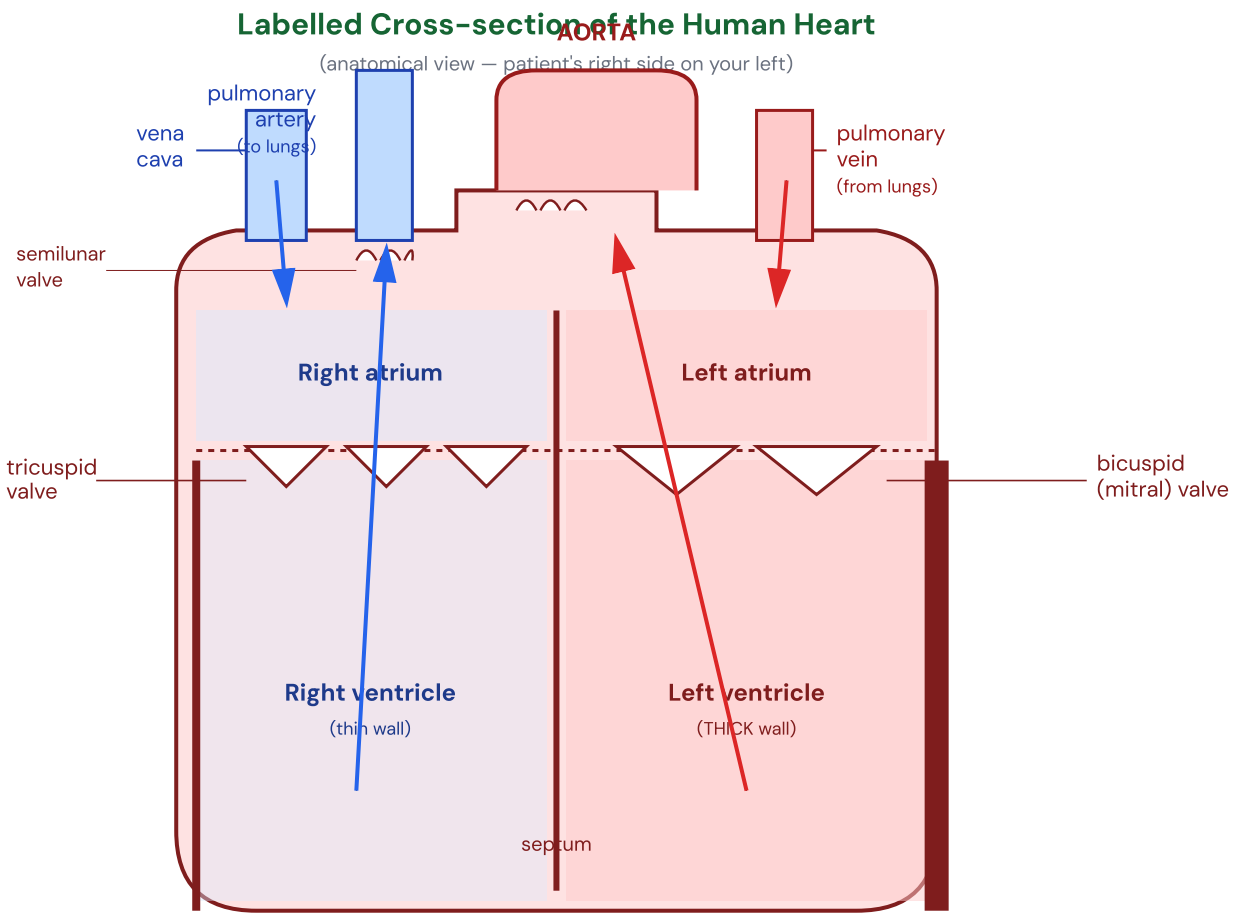
The heart is a muscular pump made of **cardiac muscle**. It has **four chambers**: two upper **atria** (thin-walled, receive blood) and two lower **ventricles** (thick-walled, pump blood out). A thick muscular wall called the **septum** separates the right and left sides, preventing oxygenated and deoxygenated blood from mixing.

Valves (one-way)

- **Tricuspid valve** — between right atrium and right ventricle (3 flaps).
- **Bicuspid (mitral) valve** — between left atrium and left ventricle (2 flaps).
- **Semilunar valves** — one in the pulmonary artery and one in the aorta, preventing backflow of blood into the ventricles after contraction.

Blood flow through the heart

```
body  →  vena cava  →  right atrium  →  tricuspid valve  →  right ventricle
        →  (semilunar valve)  →  pulmonary artery  →  LUNGS
LUNGS  →  pulmonary vein  →  left atrium  →  bicuspid valve  →  left ventricle
        →  (semilunar valve)  →  aorta  →  BODY
```



8.18 Why the Left Ventricle Wall Is Thicker

The right ventricle pumps blood only to the lungs — a short distance away. The left ventricle pumps blood *all the way around the body*, against more resistance, to reach every cell.

The left ventricle therefore needs a **much thicker, more muscular wall** so that it can generate a **higher pressure** when it contracts. The right ventricle wall is thinner because a high pressure would damage the delicate capillaries in the lungs.

> **Exam Tip:** Common mark-scheme phrase: "The left ventricle has a thicker muscular wall so that it can contract with *more force*, generating a *higher pressure* to pump blood all around the body."

8.19 Cardiac Output

> **Cardiac output** is the volume of blood pumped by the heart (by one ventricle) per minute.
> **Stroke volume** is the volume of blood pumped out per beat.
> **Heart rate** is the number of beats per minute.

```
Cardiac output (cm³ min⁻¹)  =  stroke volume (cm³)  ×  heart rate (beats min⁻¹)
```

Worked example

A person has a stroke volume of 70 cm^3 and a heart rate of 72 beats per minute. Calculate their cardiac output.

```
cardiac output = 70 × 72 = 5040 cm³ min⁻¹ (about 5 litres per minute)
```

During exercise, both the *heart rate* and *stroke volume* rise, so cardiac output can increase to 25 litres min^{-1} or more. This delivers oxygen and glucose to active muscles and removes CO_2 and lactic acid more quickly.

8.20 Treating Heart Problems

Treatment	Used for	How it works	Issues
Artificial pacemaker	Irregular heart rhythm (arrhythmia)	A small electrical device implanted under the skin sends impulses to the heart to keep it beating regularly	Needs surgery to fit; battery must be replaced after ~10 years
Stent	Narrowed coronary arteries (atherosclerosis)	A wire mesh tube inserted into the artery holds it open, restoring blood flow to the heart muscle	Risk of infection or blood clot at the site; does not treat the underlying disease
Artificial valve	Leaky or stiff heart valves	A mechanical (or animal-tissue) valve replaces the faulty one, restoring one-way flow	Patient may need long-term anticoagulant (blood-thinning) drugs; mechanical valves last longer than biological
Donor heart transplant	Severe heart failure	Diseased heart is replaced with a healthy donor heart	Shortage of donors; immunosuppressants needed to prevent rejection
Artificial heart	As a bridge while waiting for a donor	A mechanical pump replaces (or assists) the heart's function	Currently short-term only; risk of blood clots and infection

Exam Tip: When *evaluating* treatments, compare in terms of (i) effectiveness, (ii) risks/side-effects, (iii) cost and (iv) availability. Avoid vague comments like "bad" or "good" — give a specific biological reason.

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Key Terms — Topic 8

Term	Definition
SA:V ratio	Surface area divided by volume; decreases as an organism gets larger.
Exchange surface	A specialised surface across which substances pass between the body and its environment.
Alveolus	Tiny air sac in the lungs where gas exchange occurs.
Ventilation	The movement of air into and out of the lungs.
Respiration	Chemical reactions that release energy from glucose in cells.
Aerobic respiration	Respiration with oxygen; releases a large amount of energy. Produces CO_2 and H_2O.
Anaerobic respiration	Respiration without oxygen; releases a small amount of energy. Produces lactic acid (animals) or ethanol + CO_2 (yeast).
Oxygen debt	The oxygen needed after exercise to break down the lactic acid that built up during anaerobic respiration.
Haemoglobin	Iron-containing protein in red blood cells that binds oxygen reversibly.
Plasma	Straw-coloured liquid component of blood; transports cells, nutrients, CO_2, urea and hormones.
Artery / Vein / Capillary	Artery: carries blood away from heart (thick wall, narrow lumen). Vein: returns blood to heart (thin wall, wide lumen, valves). Capillary: exchange vessel (one cell thick).
Double circulation	Blood passes through the heart twice per full circuit: once to the lungs, once to the body.
Atrium / Ventricle	Atrium = upper, thin-walled chamber receiving blood. Ventricle = lower, thick-walled chamber pumping blood out.

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Cardiac output	Volume of blood pumped per minute = stroke volume × heart rate.
Pacemaker	Natural cluster of cells (SAN) or artificial electrical device that controls heartbeat rhythm.
Stent	Mesh tube used to keep a narrowed artery open.

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